SAMPLE

# Cosmos Unveiled: Big Bang Nucleosynthesis

The First Elements

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## Introduction

Big Bang nucleosynthesis is the process by which the light elements, primarily hydrogen, helium, and lithium, were produced in the first few minutes after the Big Bang. This process, which occurred when the universe was about 1 second to 3 minutes old, is one of the three pillars of the Big Bang theory, along with the expansion of the universe and the cosmic microwave background.

The theory of Big Bang nucleosynthesis was developed by George Gamow, Ralph Alpher, and Robert Herman in the 1940s and 1950s. They calculated that the hot, dense early universe should have produced specific amounts of the light elements, and they predicted that the universe should be about 75 percent hydrogen and 25 percent helium by mass.

The observed abundances of these elements, measured through spectroscopic observations of stars and gas clouds, match these predictions with remarkable precision, providing one of the strongest tests of the Big Bang theory. The key parameter in Big Bang nucleosynthesis is the baryon density, which is the density of ordinary matter (protons and neutrons) in the universe. The baryon density determines the rate of the nuclear reactions that produce the light elements, and it affects the final abundances.

The baryon density measured by the Planck satellite through the cosmic microwave background is consistent with the baryon density inferred from the observed abundances of the light elements, providing a remarkable confirmation of the consistency of the standard cosmological model. The process of Big Bang nucleosynthesis begins about 1 second after the Big Bang, when the temperature of the universe falls below about 10^10 Kelvin.

At this temperature, the neutrons and protons are no longer in thermal equilibrium, and the ratio of neutrons to protons is fixed at about 1:7. About 100 seconds later, when the temperature falls to about 10^9 Kelvin, protons and neutrons begin to combine to form deuterium and helium-4.

By the end of nucleosynthesis, about 3 minutes after the Big Bang, about 25 percent of the mass of the universe is in helium-4 and 75 percent is in hydrogen, with trace amounts of deuterium, helium-3, and lithium-7. The importance of Big Bang nucleosynthesis extends far beyond the confirmation of the Big Bang theory. BBN provides the most precise measurement of the baryon density of the universe, the density of ordinary matter that makes up stars, planets, and living things.

BBN also constrains the number of neutrino species, the neutron half-life, and other fundamental parameters of particle physics. The theory is a unique probe of the conditions in the first few minutes after the Big Bang, providing information about the early universe that cannot be obtained by any other method. The development of the BBN theory has been one of the great success stories of modern astrophysics.

The original calculations by Gamow, Alpher, and Herman in the 1940s were remarkably prescient, predicting the correct helium abundance and the existence of the CMB before either was observed. The subsequent refinement of the theory, with improved nuclear reaction rates and computing power, has improved the precision of the predictions by more than an order of magnitude.

The comparison of the BBN predictions with the observed abundances is one of the strongest tests of the Big Bang theory and one of the most precise tests in all of cosmology. The baryon density determined by BBN is one of the most important parameters in cosmology. It determines the amount of ordinary matter in the universe, which is about 5 percent of the total mass-energy content. The remaining 95 percent is dark matter and dark energy, which do not participate in nuclear reactions.

The fact that BBN can determine the baryon density to a precision of about 5 percent from the abundances of the light elements, and that this value agrees with the CMB determination to the same precision, is a remarkable achievement. The Big Bang nucleosynthesis theory is one of the most precisely tested theories in all of science.

The predictions of the theory, which are based on the physics of the first few minutes after the Big Bang, have been confirmed by observations with precision of about 5 percent for deuterium and about 10 percent for helium-4. The agreement between the predictions and the observations is a powerful confirmation of the Big Bang origin of the universe and a testament to the power of theoretical physics and modern astronomy.

The baryon density, which is the density of ordinary matter, is about 5 percent of the total mass-energy content of the universe. The Big Bang nucleosynthesis theory has been tested against observations for over 70 years, and it has passed every test.

The predicted abundances of the light elements match the observed abundances with remarkable precision, providing one of the strongest confirmations of the Big Bang theory. The BBN theory is also important because it connects cosmology to particle physics. The BBN predictions depend on the properties of fundamental particles, including the neutron half-life, the number of neutrino species, and the baryon density.

The observed abundances constrain these fundamental parameters, making BBN a unique probe of particle physics in the early universe. The BBN theory has been refined and improved over the decades since its initial development. The modern BBN codes, including PArthENoPE and AlterBBN, calculate the abundances with precision of about 0.1 percent.

BBN is one of the three pillars of the Big Bang theory. The BBN theory connects cosmology to particle physics in a unique and powerful way. The study of BBN will continue to be at the frontier of science. BBN remains a cornerstone of modern cosmology. This is cosmology.

In physical cosmology, Big Bang nucleosynthesis (also known as primordial nucleosynthesis, and abbreviated as BBN) is a model for the production of light nuclei (including, but not limited to, 2H, 3He, 4He, and 7Li) during the first ~20 minutes of the evolution of the universe. The predictions of BBN rely on a combination of thermodynamic arguments and measurements of rates of nuclear reactions that are expected to be active in the early universe.

Refined models tend to agree well with observations of these light element abundances. The model is one of the key concepts in standard cosmology. Elements heavier than lithium are instead created in appreciable amounts at later times via stellar nucleosynthesis, through the formation, evolution and death of stars. The history of Big Bang nucleosynthesis research began with George Gamow, a nuclear physicist who thought that Victor Goldschmidt's data on the abundance of elements in the universe might be predicted by nuclear reactions.

Early in the 1940s Gamow realized that very high temperatures and pressures needed for nuclear reactions implied an explosion followed by expansion. Calculations by his student Ralph Alpher, published in the famous Alpher-Bethe-Gamow paper, outlined an early theory of light-element production in the early universe. These early efforts did not involve specific nuclear reactions. When Enrico Fermi and Anthony L. Turkevich tried, they found they would predict hydrogen and helium, but no higher elements.

The key problem was a "mass gap": there are no nuclei with masses of 5 and 8 atomic mass units needed for the reactions to reach higher masses. In 1953 this problem seemed insurmountable and cast doubt on the entire concept. The competing cosmological model, the steady state model, appearing in two 1948 publications by Fred Hoyle and by Herman Bondi and Thomas Gold, had no mass-gap issue because it ignored nuclear processes initially.

## Chapter 01: The Physics of the Early Universe

The physics of the early universe, in the first few minutes after the Big Bang, is described by a combination of general relativity, thermodynamics, and nuclear physics. The universe was extremely hot and dense, with temperatures above 10^9 Kelvin, and it was filled with a plasma of protons, neutrons, electrons, photons, and neutrinos.

In the first second after the Big Bang, the temperature was above 10^10 Kelvin, and protons and neutrons were in thermal equilibrium through weak nuclear reactions. The reactions that maintained the equilibrium were the conversion of protons to neutrons and vice versa, through interactions with electrons, positrons, and neutrinos. The equilibrium ratio of neutrons to protons depended on the temperature, with equal numbers at very high temperatures and a decreasing neutron fraction as the temperature fell.

When the temperature fell below about 10^10 Kelvin, about 1 second after the Big Bang, the weak nuclear reactions became too slow to maintain the equilibrium, and the neutron-proton ratio was frozen at about 1:7. This ratio is determined by the mass difference between the neutron and the proton, which is about 1.293 MeV, and by the temperature at which the reactions freeze out.

After freeze-out, the neutrons begin to decay into protons with a half-life of about 880 seconds, slightly reducing the neutron fraction before nucleosynthesis begins. The freeze-out of the weak interactions is a crucial moment in the history of the universe. Before freeze-out, the neutron-proton ratio was determined by thermal equilibrium. After freeze-out, the ratio is determined by the decay of neutrons and by the nuclear reactions that incorporate neutrons into nuclei.

The freeze-out temperature, which depends on the expansion rate of the universe, is sensitive to the number of

neutrino species. If there are more neutrino species, the expansion rate is faster, the freeze-out occurs earlier, and the neutron-proton ratio is higher, leading to more helium-4. The physics of the early universe is described by the standard model of particle physics and by general relativity.

The standard model describes the fundamental particles and their interactions, including the quarks and leptons that make up protons and neutrons, and the weak, electromagnetic, and strong forces that govern their interactions. General relativity describes the expansion of the universe and the gravitational dynamics. The combination of these two theories, applied to the conditions of the early universe, gives the predictions of BBN.

Where the weak interactions finally release their grip hinges on how briskly space dilutes the radiation bath, so the exact freeze-out temperature listens both to how many light species populate that bath and to whatever exotic physics might quicken the dilution. The sensitivity of BBN to the number of neutrino species is one of the most important features of the theory.

Every extra light species dilutes the radiation faster, so the weak reactions switch off sooner, neutrons linger in larger numbers, and the final helium crop grows richer. The observed helium-4 abundance constrains the number of neutrino species to be close to 3, the value in the standard model. This constraint was derived from BBN before the LEP collider directly measured the number of neutrino species in 1989, and it is one of the great successes of the theory.

The physics of the early universe is remarkably well understood, despite the extreme conditions. The temperature, density, and composition of the universe at the time of BBN can be calculated with high precision from the standard model of particle physics and general relativity. The nuclear reactions that produce the light elements are described by nuclear physics, and the reaction rates have been measured in the laboratory.

The combination of these well-understood physical processes gives the BBN predictions, which are among the most precise predictions in cosmology. The freeze-out of the weak interactions is a key moment in the history of the universe. The BBN constraint on the number of neutrino species, derived from the helium-4 abundance, is one of the most important results in cosmology.

The freeze-out of the weak interactions is one of the key moments in the history of the universe. The freeze-out occurs when the weak nuclear reactions become too slow to maintain the thermal equilibrium between neutrons and protons.

The freeze-out temperature, which is about 10^10 Kelvin, depends on the expansion rate of the universe, which in turn depends on the number of neutrino species and the total radiation density. The physics of the early universe is remarkably well understood. The freeze-out of the weak interactions determines the neutron-proton ratio.

In 1957, Hoyle teamed up with Margaret Burbidge, Geoffrey Burbidge, and William A. Fowler to develop a complete account of how stars forge the heavier elements, the landmark synthesis known as B2FH.

That theory matched the abundance of almost all of the elements except for hydrogen and helium, the elements correctly predicted by the Big Bang theory.

The discovery of the cosmic microwave background radiation in 1965 strongly supported the Big Bang theory and it also allowed Jim Peebles to use the estimated background temperature, 3K, in the first detailed calculations of the primordial isotopic abundances, showing a He abundance between 26 and 28%. The following year Hoyle, Fowler, and Robert Wagoner showed that the very light elements like He require higher temperatures than stars provide, evidence that most scientists took as support for the Big Bang model.

In 1972 the Copernicus satellite began to provide accurate measurements of deuterium abundance. Since primordial synthesis of deuterium depends more strongly on baryon density than does synthesis of 3He, the deuterium abundance could be used to infer primordial density. By the 1990s additional measurements concluded that this density implied there must be mass other than baryons in the universe, an idea that came to be called dark matter.

These nucleosynthesis methods have been refined over the years using updated estimates of the input nuclear reaction rates. The first systematic Monte Carlo study of how nuclear reaction rate uncertainties impact isotope predictions, over the relevant temperature range, was carried out in 1993. Big Bang nucleosynthesis describes the production of atomic nuclei from protons and neutrons in the expanding and cooling early universe. The process occurs at equilibrium with protons and neutrons combining to create nuclei and the nuclei disintegrating.

Nobody had to guess the starting conditions, because thermodynamics fixes them. A second after the beginning,

the temperature sat near ten billion degrees, hot enough that radiation outweighed matter by more than a million to one. Under those conditions, neutrons and protons were not yet a fixed inventory but a bubbling market, flipping back and forth through the weak interaction at a feverish rate. The bookkeeping that governs this market is the Saha equation, which compares the rates of the flipping reactions against the pace at which the cosmos dilutes.

The clock matters as much as the heat. Between one second and three minutes the radiation temperature slid from about ten billion degrees to roughly a billion, and each factor of ten in that slide switched off one class of reaction. A single takeaway serves for all of it: nucleosynthesis is a race between reaction speeds and cosmic dilution, and every product that survives is whatever the race left unfinished when the track ran out.